

INVERSE TRIGONOMETRY – LEVEL 1

SECTION 1: BASIC DEFINITIONS & PRINCIPAL VALUES (Q1–Q20)

1. An inverse trigonometric function reverses a trigonometric function after domain restriction.
 2. Principal value is the unique value of an inverse function in its defined range.
 3. $y = \sin^{-1} x \Rightarrow \sin y = x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
 4. $y = \cos^{-1} x \Rightarrow \cos y = x, y \in [0, \pi]$
 5. $y = \tan^{-1} x \Rightarrow \tan y = x, y \in (-\frac{\pi}{2}, \frac{\pi}{2})$
 6. Domain: $[-1, 1]$, Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$
 7. Domain: $[-1, 1]$, Range: $[0, \pi]$
 8. Domain: $\mathbb{R}$, Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$
 9. $[-\frac{\pi}{2}, \frac{\pi}{2}]$
 10. $[0, \pi]$
 11. $(-\frac{\pi}{2}, \frac{\pi}{2})$
 12. $\cot^{-1} x : (0, \pi), \sec^{-1} x : [0, \pi] \setminus \{\frac{\pi}{2}\}, \csc^{-1} x : [-\frac{\pi}{2}, \frac{\pi}{2}] \setminus \{0\}$
 13. Because $\sin x$ is one-one and onto on that interval.
 14. Because $\cos x$ is one-one on $[0, \pi]$.
 15. To make trigonometric functions invertible.
 16. $[-\frac{\pi}{2}, \frac{\pi}{2}]$
 17. $[0, \pi]$
 18. $(-\frac{\pi}{2}, \frac{\pi}{2})$
 19. $\sin^{-1}(\sin \frac{3\pi}{4}) = \frac{\pi}{4} \neq \frac{3\pi}{4}$
 20. $\cos^{-1}(\cos \frac{5\pi}{3}) = \frac{\pi}{3} \neq \frac{5\pi}{3}$
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SECTION 2: SIMPLE EVALUATION (Q21–Q50)

21. 0
22. $\frac{\pi}{6}$
23. $\frac{\pi}{3}$
24. $-\frac{\pi}{6}$
25. 0
26. $\frac{\pi}{3}$
27. $\frac{2\pi}{3}$
28. $\frac{\pi}{2}$
29. 0
30. $\frac{\pi}{4}$
31. $\frac{\pi}{3}$

32. $-\frac{\pi}{4}$
33. $\frac{\pi}{4}$
34. $\frac{\pi}{3}$
35. $\frac{\pi}{6}$
36. $\frac{\pi}{4}$
37. $\frac{\pi}{3}$
38. $-\frac{3\pi}{4}$
39. $\frac{\pi}{3}$
40. $\frac{\pi}{6}$
41. $\frac{3}{5}$
42. $\frac{4}{5}$
43. $\frac{7}{3}$
44. $\pi - 5$
45. $4 - \pi$
46. -2
47. 0
48. $\frac{\pi}{2}$
49. $\frac{\pi}{12}$
50. $\frac{\pi}{2}$

SECTION 3: IDENTITIES & PROOFS (Q51–Q80)

51. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
52. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
53. $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$
54. $\sin^{-1}(-x) = -\sin^{-1} x$
55. $\tan^{-1}(-x) = -\tan^{-1} x$
56. $\cos^{-1}(-x) = \pi - \cos^{-1} x$
57. $\cot^{-1}(-x) = \pi - \cot^{-1} x$
58. $\sin^{-1} x = \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right)$
59. $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$
60. $2 \sin^{-1} x = \cos^{-1}(1 - 2x^2)$
61. $x^2 + y^2 = 1$
62. Proven
63. Proven
64. Proven
65. Proven

66. $\frac{\pi}{4}$
 67. $\frac{\pi}{2}$
 68. $\frac{\pi}{2}$
 69. $\frac{\pi}{2}$
 70. $\frac{\pi}{2}$
 71. $\frac{\pi}{2}$
 72. $\frac{\pi}{2}$
 73. Constant = $\frac{\pi}{2}$
 74. Odd
 75. Neither
 76. Odd
 77. π
 78. $x^2 + y^2 = 1$
 79. $x = y$
 80. $x = y$
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SECTION 4: COMPOSITION (Q81–Q110)

81. x
82. x
83. x
84. $\sqrt{1-x^2}$
85. $\sqrt{1-x^2}$
86. $\frac{x}{\sqrt{1-x^2}}$
87. $\frac{x}{\sqrt{1+x^2}}$
88. $\frac{1}{\sqrt{1+x^2}}$
89. $\frac{1}{x}$
90. $\frac{4}{5}$
91. $\frac{3}{5}$
92. $\frac{12}{5}$
93. $\frac{3}{5}$
94. $\frac{4}{5}$
95. $\frac{13}{12}$
96. $\frac{13}{12}$
97. $2 \tan^{-1} t$
98. $2 \tan^{-1} t$
99. $2 \tan^{-1} x$
100. $\sin^{-1}(\sin x)$

- 101. $\cos^{-1}(\cos x)$
 - 102. $2x$
 - 103. $2x$
 - 104. $\cos x$
 - 105. $\sin x$
 - 106. $\sin^{-1}(\sin 2x)$
 - 107. $\cos^{-1}(\cos 2x)$
 - 108. $\frac{2x}{1+x^2}$
 - 109. $\frac{1-x^2}{1+x^2}$
 - 110. $1 - 2x^2$
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SECTION 5: EQUATIONS & INEQUALITIES (Q111–Q150)

- 111. $x = \frac{1}{2}$
- 112. $x = -\frac{1}{2}$
- 113. $x = 1$
- 114. $x = \frac{1}{2}$
- 115. $x = \frac{1}{\sqrt{2}}$
- 116. $x = 0$
- 117. $[-1, \frac{1}{\sqrt{2}}]$
- 118. $[-1, \frac{1}{2}]$
- 119. $x = \tan \frac{\pi}{8}$
- 120. $x = \frac{1}{4}$
- 121. $x = \pm \frac{1}{2}$
- 122. $x = \sqrt{2} - 1$
- 123. $x = \frac{3}{5}$
- 124. $x = \frac{4}{5}$
- 125. $x = \frac{7}{3}$
- 126. $x \in [-1, 1]$
- 127. $x = \pm \frac{1}{\sqrt{2}}$
- 128. $x = \pm \frac{1}{\sqrt{2}}$
- 129. $x = \frac{1}{2}$
- 130. $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- 131. $[0, \pi]$
- 132. $(-\frac{\pi}{2}, \frac{\pi}{2})$
- 133. $[-1, 1]$

134. $\mathbb{R}$

135. $\mathbb{R}$

136. $x > 0$

137. $x > 0$

138. $x \geq 0$

139. $\frac{1}{2} \leq x \leq 1$

140. $[0, 1]$

141. $x = \frac{1}{\sqrt{2}}$

142. $x = \frac{1}{2}$

143. $x = \frac{1}{2}$

144. $x = 0$

145. $x = \frac{1}{\sqrt{2}}$

146. $[-1, 1]$

147. $[-1, 1]$

148. Continuous

149. Continuous

150. Strictly increasing on $\mathbb{R}$